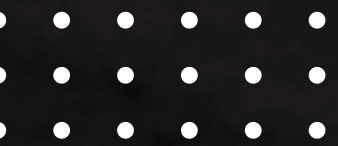




EVOASTRA VENTURES -
MINI PROJECT



WEB SCRAPING TATA **CARS**

Data Scraping and Analysis of Tata Cars from Cars24

Presented by
TEAM - C

Date
**September 14th,
2025**



OUR TEAM

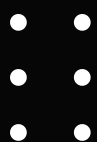


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OVERVIEW



This project involved web scraping Tata car listings from Cars24 using Selenium to extract details such as year, fuel type, transmission, price, and ownership. After cleaning and analyzing the dataset, we generated key insights through visualizations. The analysis revealed that most resale cars are recent models, predominantly Petrol and Manual, with first-owner cars dominating. Prices largely fall between ₹5–12 lakhs, and newer cars retain higher value. Overall, the project demonstrates how data scraping and analytics can uncover meaningful market trends and support better decision-making.

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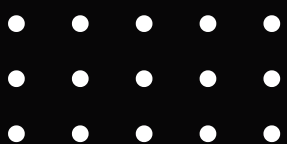
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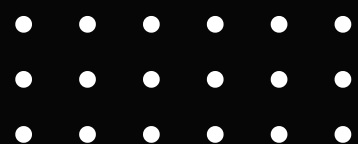


ABOUT COMPANY



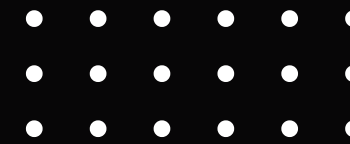
EvoAstra Ventures is a dynamic company specializing in Data Science and AI services, delivering customized solutions, AI strategy consulting, and advanced machine learning model development.

- Founded in 2023 by Aniket Manwatkar and Syeda Mariya Quadri.
- Successfully completed over 350 projects across industries.
- Currently serving 16+ leading brands and organizations with innovative AI-driven solutions.





PROJECT



OBJECTIVE

- Collect, clean, and analyze Tata car listings from Cars24 using Selenium and Python.
- Identify trends in cars:
 1. Variant
 2. Kilometers Driven
 3. Year of Manufacture
 4. Fuel Type
 5. Transmission
 6. Price Ranges
 7. Location
 8. Number of Owners

REQUIREMENTS

- Python 3
- Selenium
- BeautifulSoup4
- WebDriver Manager
- Pandas
- OpenPyXL





METHODOLOGY



Research & Planning

- Studied the Cars24 website structure to understand how car listings are loaded dynamically.
- Identified relevant HTML tags, CSS selectors, and attributes for extracting data (e.g., car title, year, fuel type, price, and ownership).



Data Extraction (Web Scraping)

- Used Selenium WebDriver to automate browser actions and handle Cars24's infinite scroll mechanism.
- Implemented a custom scroll-and-collect function to capture all available Tata car listings in the Mumbai , Hyderabad, New Delhi region.



Data Cleaning & Transformation

- Converted Price, Kilometers Driven, and Number of Owners into numeric formats for consistency.
- Standardized text fields (e.g., Petrol vs PETROL) to avoid duplicates.
- Handled missing or incomplete data using placeholder values like N/A.
- Removed duplicate car links to ensure uniqueness of records.

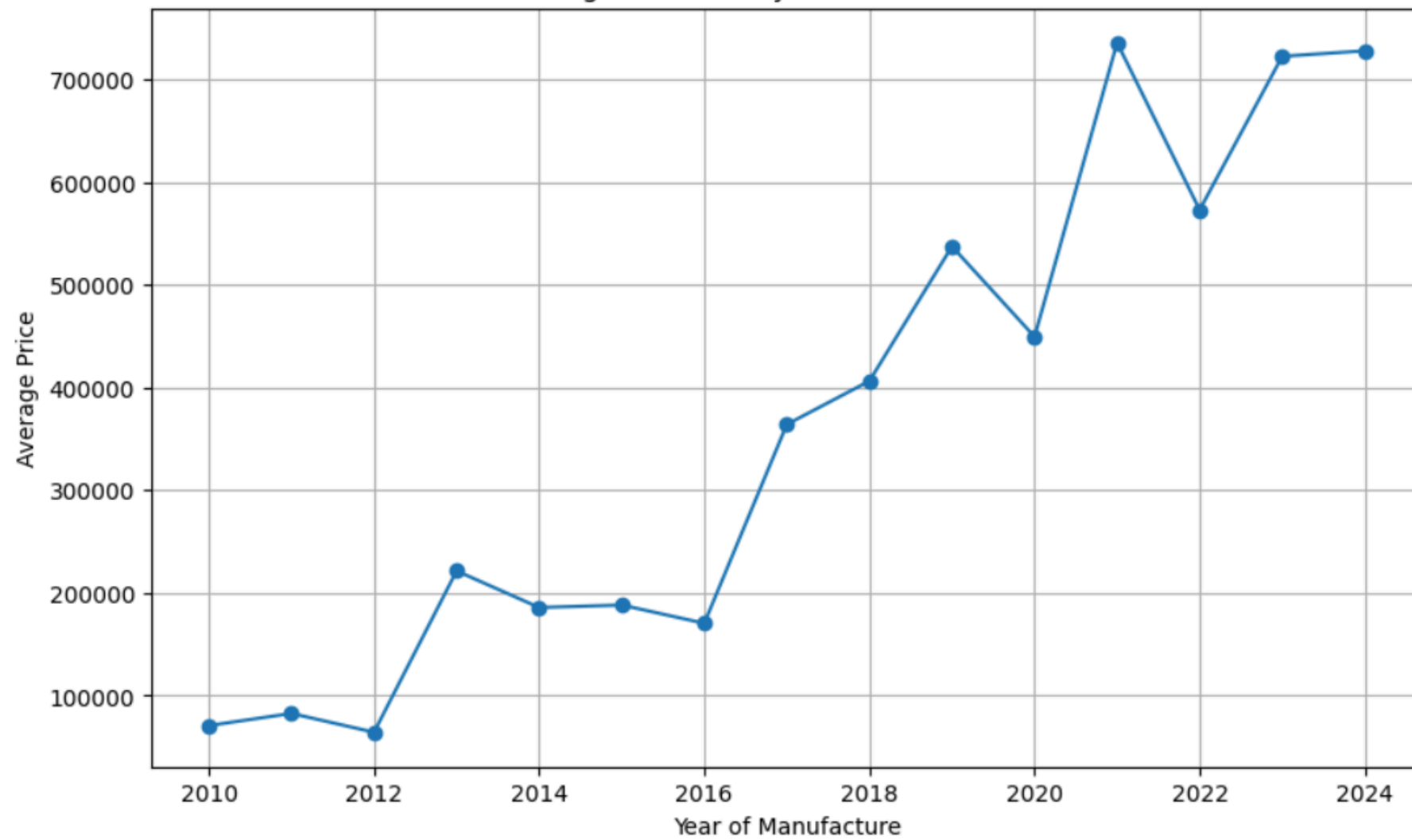




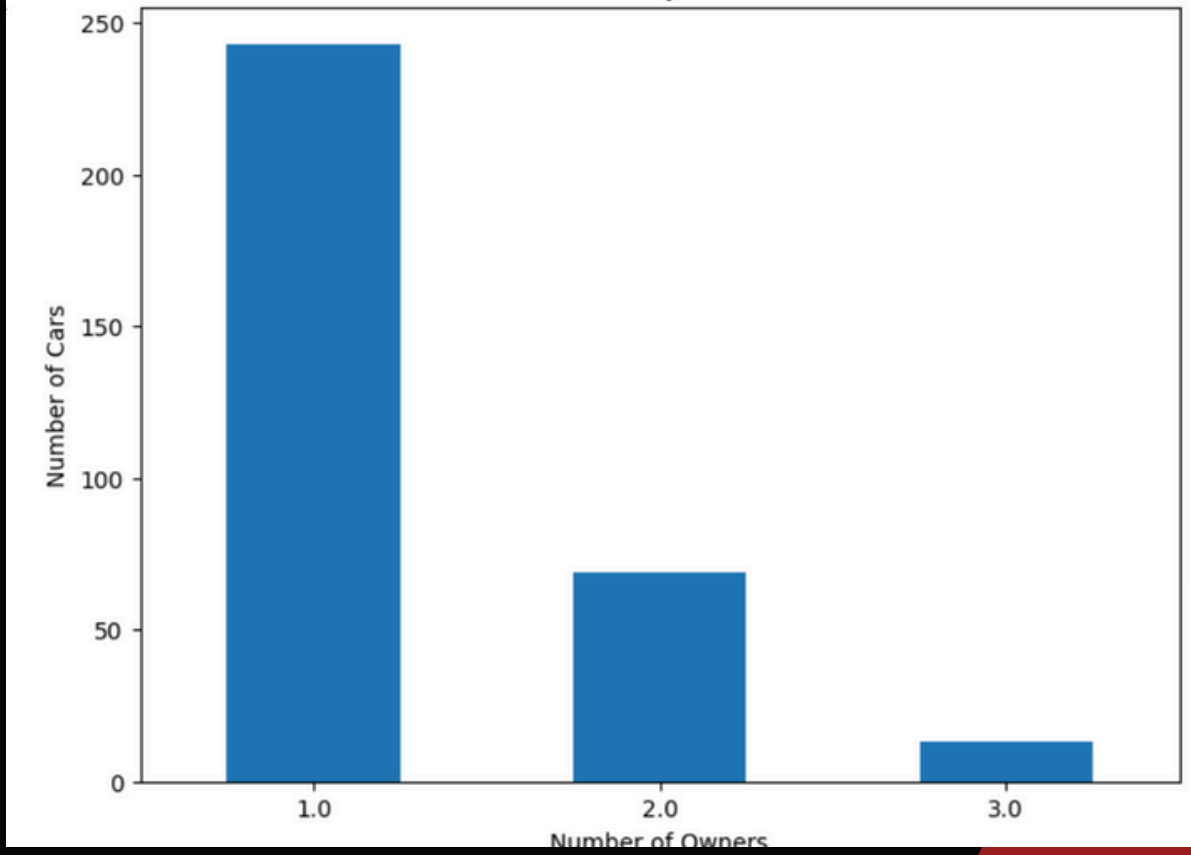
DATA VISUALIZATION



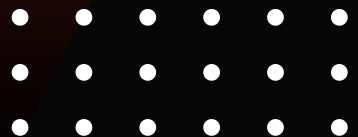
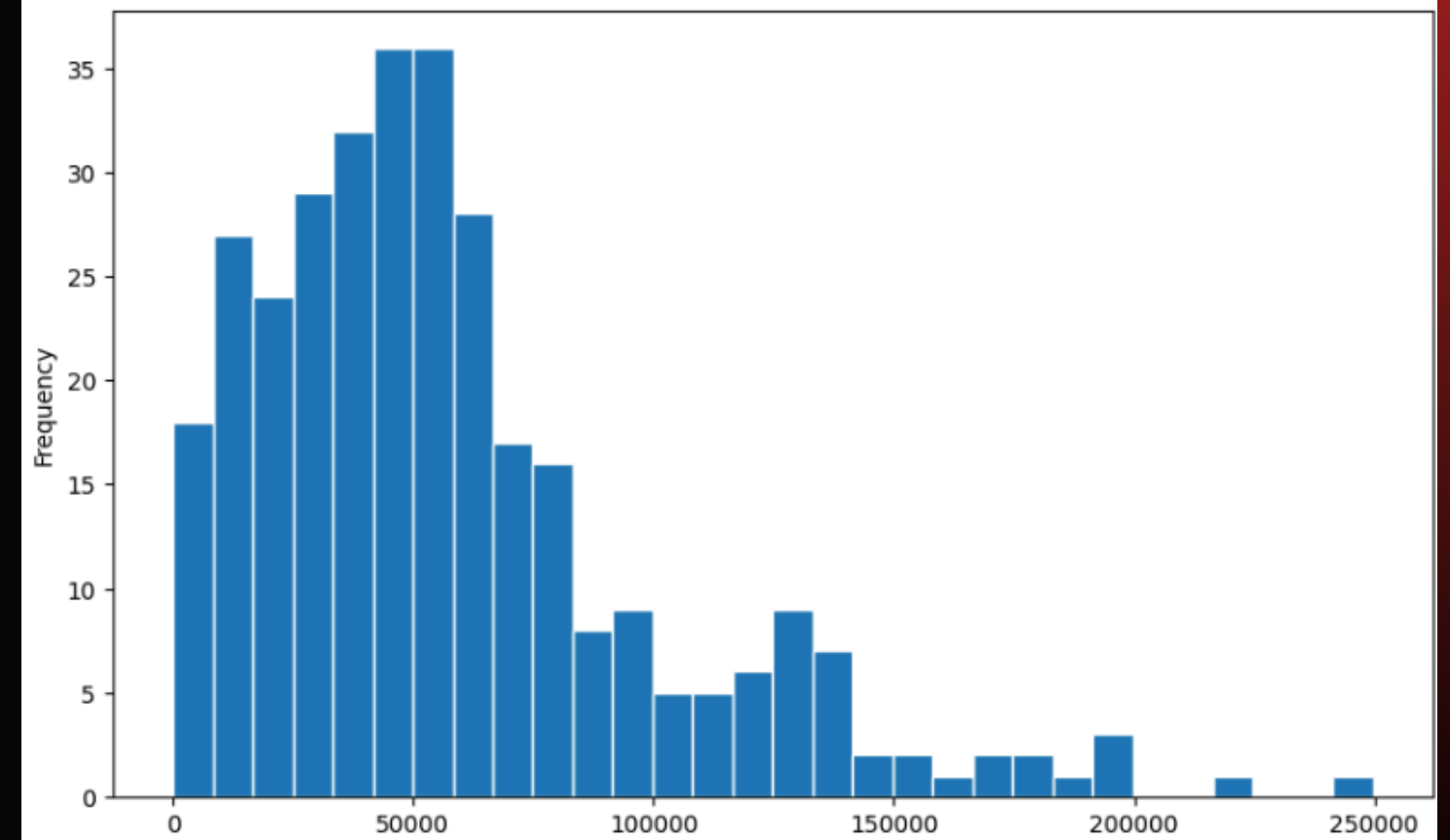
Average Car Price by Year of Manufacture



Distribution of Cars by Number of Owners



Distribution of Kilometers Driven

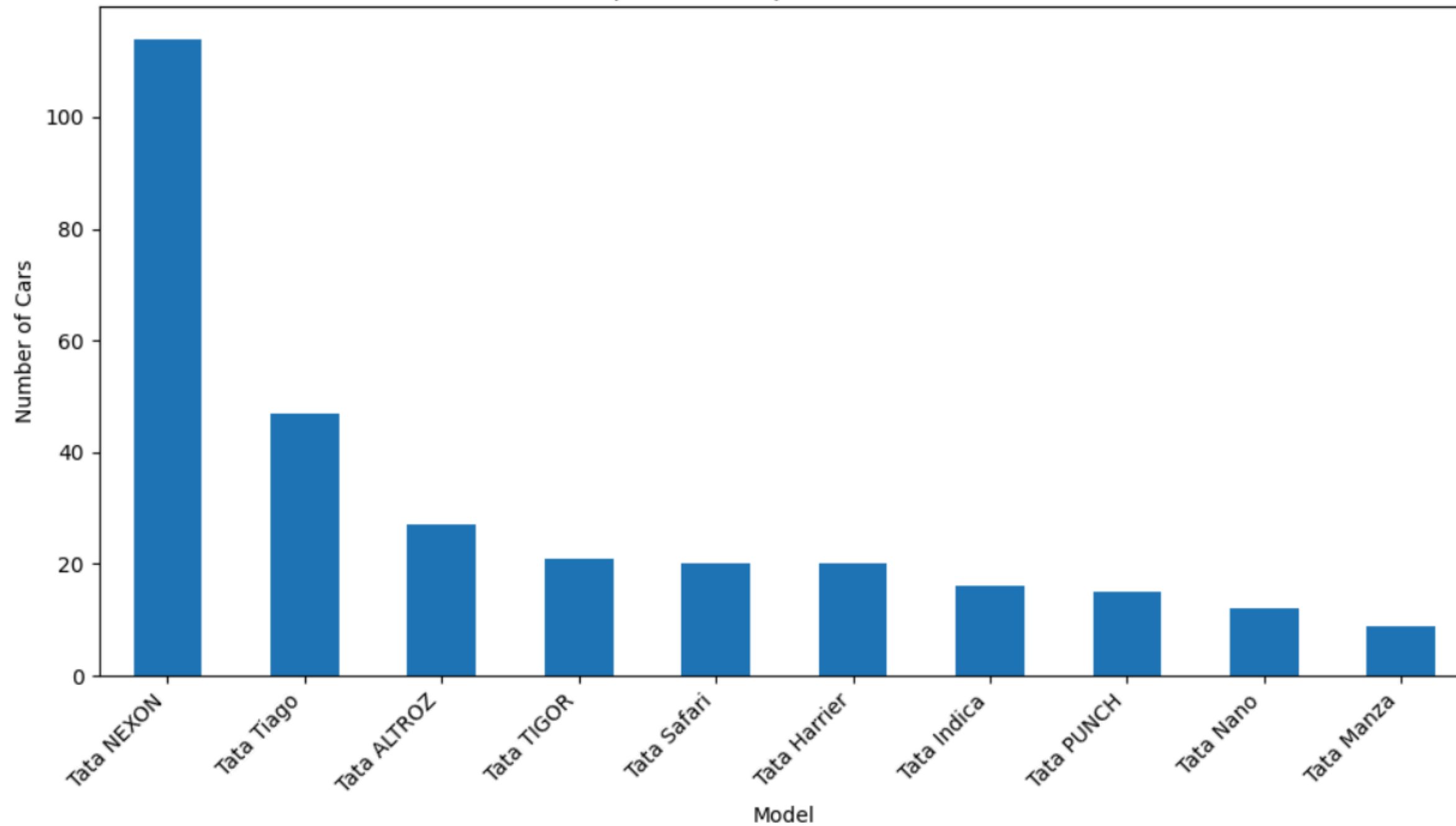




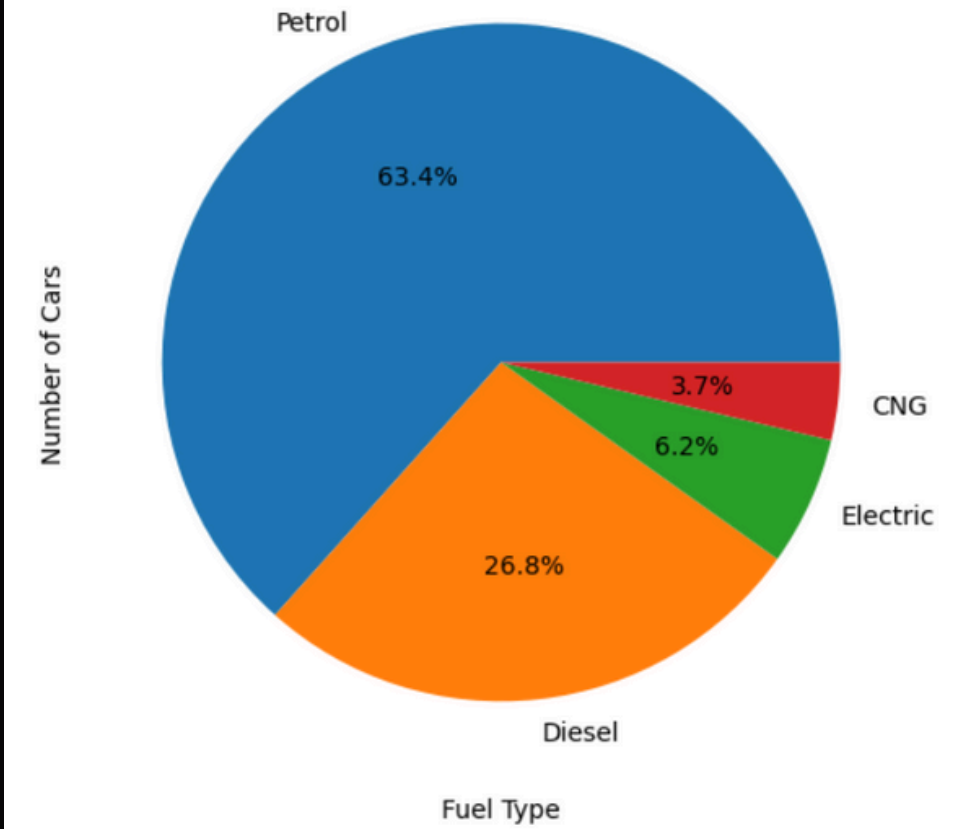
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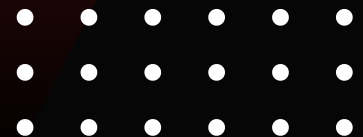
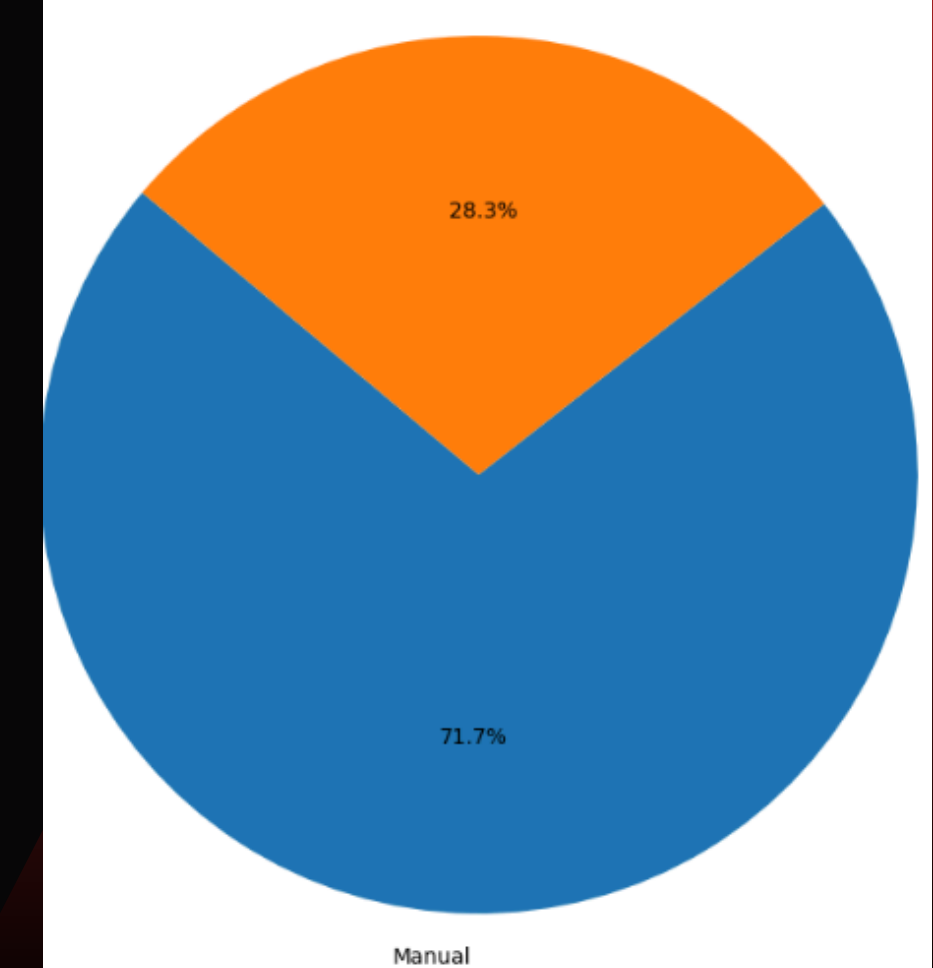
Top 10 Most Popular Car Models



Distribution of Cars by Fuel Type

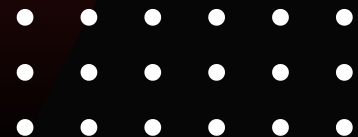
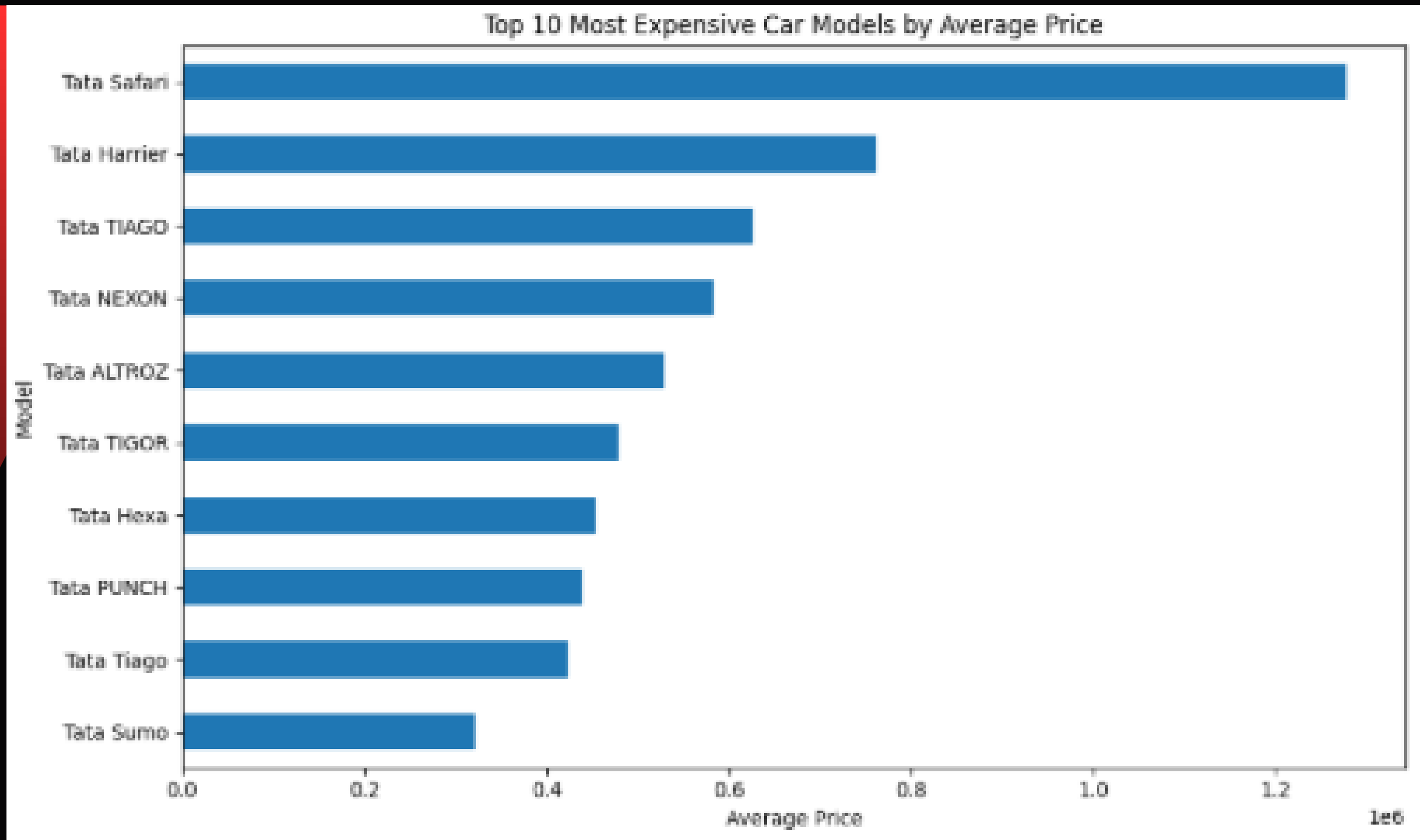


Distribution of Cars by Transmission Type





DATA VISUALIZATION

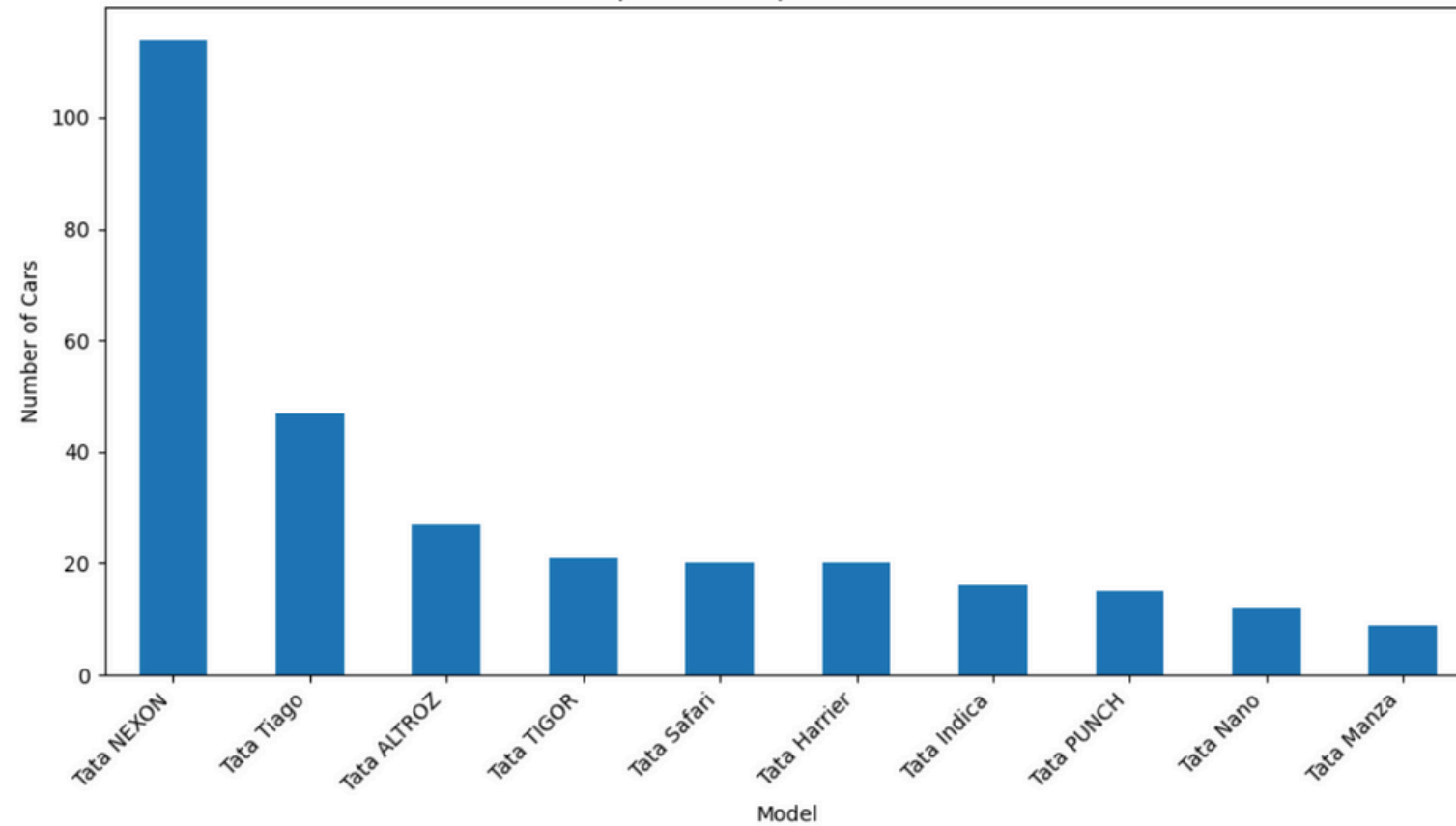




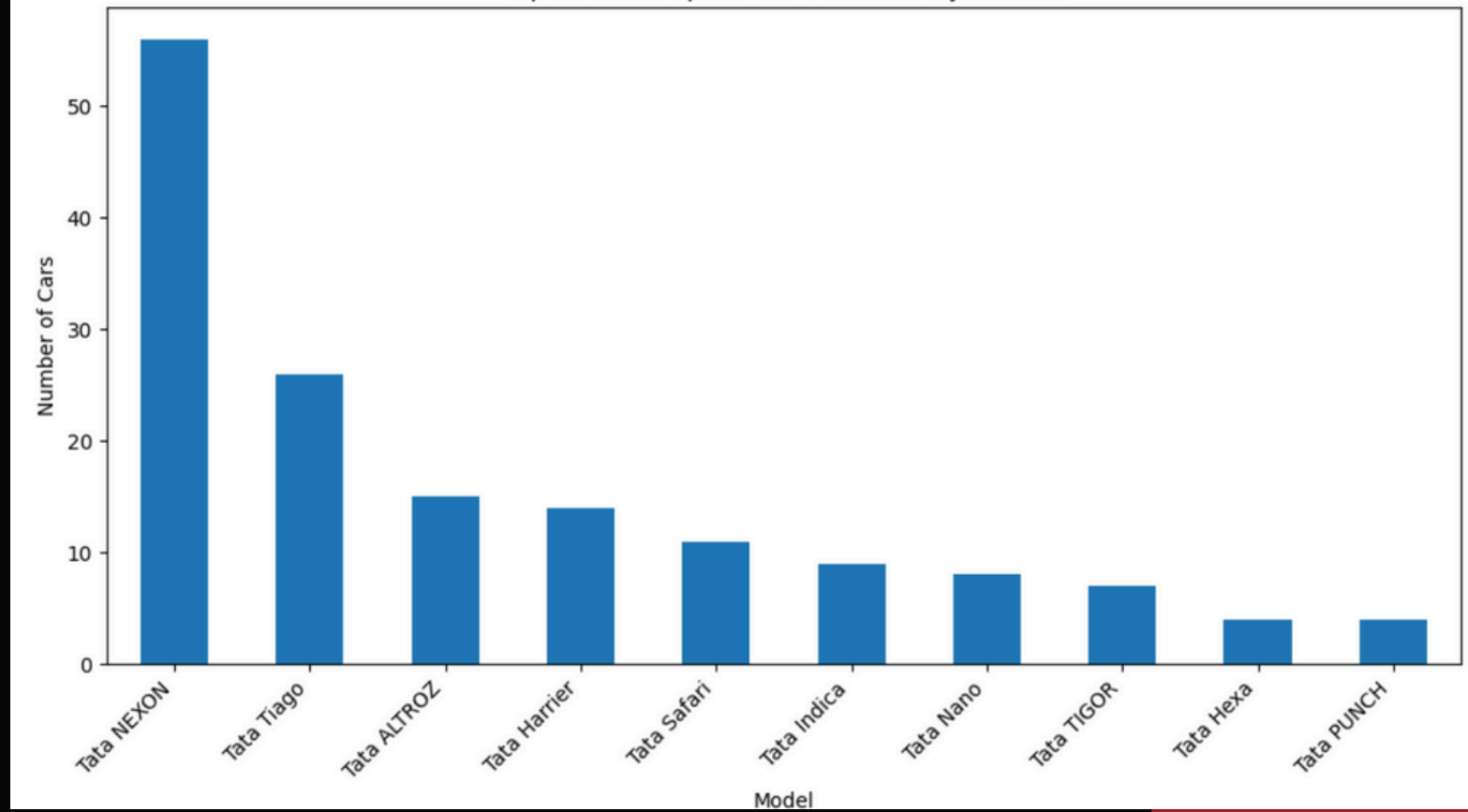
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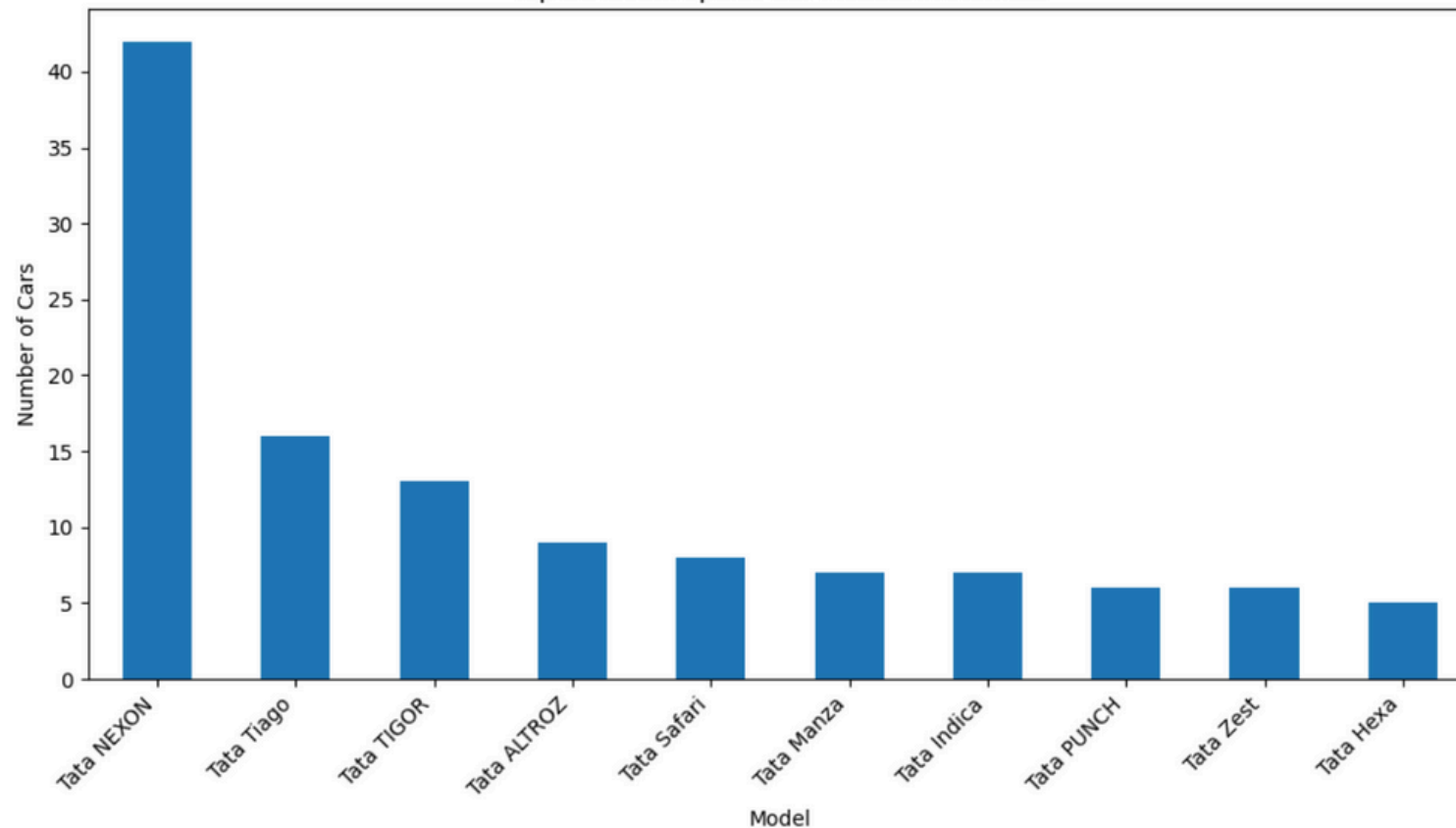
Top 10 Most Popular Car Models



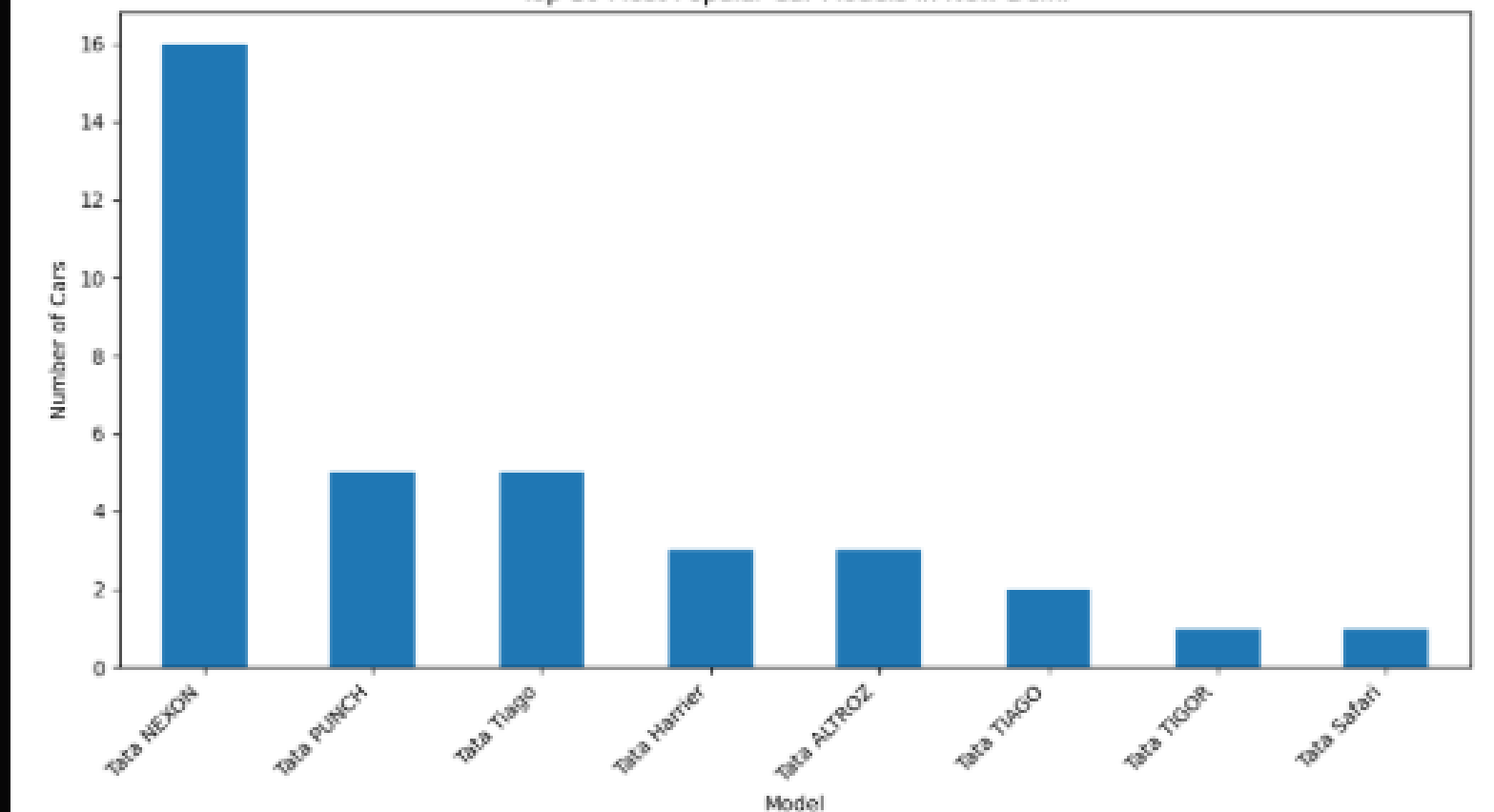
Top 10 Most Popular Car Models in Hyderabad



Top 10 Most Popular Car Models in Mumbai



Top 10 Most Popular Car Models in New Delhi





CHALLENGES & SOLUTIONS



1. Dynamic Web Content (Infinite Scroll)

- Challenge: Cars24 loads listings dynamically with infinite scroll, making static scraping ineffective.
- Solution: Implemented **Selenium** automation with a custom scroll-and-collect function to continuously load and capture car data.

2. Data Cleaning & Standardization

- Challenge: Extracted data such as **prices, kilometers, and ownership** details appeared in text format with symbols (₹, km, etc.) and inconsistencies.
- Solution: Used **Pandas** for cleaning — converted fields into numeric values, handled missing data with placeholders, and standardized categorical fields (e.g., Fuel Type, Transmission).

3. Duplicate & Redundant Data

- Challenge: Duplicate car links appeared due to repeated scroll loads, inflating dataset size.
- Solution: Removed duplicates by applying **dictionary-based link** filtering and ensuring only unique records were saved.





KEY INSIGHTS

- Most cars are from **2018 onwards** → newer models dominate resale.
- **Petrol cars dominate**; Diesel second; CNG/Electric rare.
- **Manual cars majority**, though Automatics are rising.
- **First-owner cars** dominate → better resale reliability.
- Prices mainly **₹5–12 lakhs**, with some premium outliers.
- Most cars **<60,000 km driven** → good condition.
- Listings concentrated in:
 1. **Hyderabad** → 163 listings
 2. **Mumbai** → 129 listings
 3. **New Delhi** → 36 listings
- **Newer cars (2020+)** retain higher prices → slower depreciation.





RECOMMENDATIONS & CONCLUSION



Recommendations

- Expand scraping to multiple cities for pan-India analysis.
- Develop predictive models for price estimation.
- Track customer preferences (e.g., demand for Automatics).
- Use dashboards (PowerBI/Tableau) for interactive analysis.
- Extend scraping to other brands for competitive benchmarking.



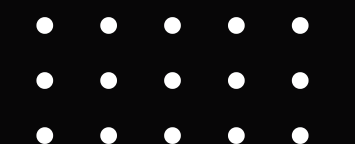
Conclusion

- Successfully scraped and analyzed Tata cars from Cars24.
- Identified resale patterns in year, fuel, transmission, ownership, and price.



Future Scope:

- Expand dataset across cities and brands.
- Apply predictive analytics for price forecasting.
- Build interactive dashboards for deeper insights.





TEAM - C



THANKYOU

